DO NOW — RECALL

Lesson 117 · Waterfall and Agile Development

Name: _____ Class: _____ Date: _____

1. What is the difference between linker and loader?

2. What is the time complexity of sequential search in the worst case?

3. After pushing A then B onto a stack, which is popped first?

4. Why must a binary search operate on ordered data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Linker resolves modules/symbols; loader maps executable into memory to run.

2. Model answer

O(n).

3. Model answer

B, because a stack is LIFO.

4. Model answer

It discards half the search space using an ordering comparison.

By the end, you can…

• Sequential analysis, design, implementation, testing and maintenance.

• Iterative delivery, feedback and reprioritisation in short cycles.

• Stability, regulation, stakeholder access and change risk affect suitability.

Vocabulary — know these terms

waterfall — Stage-based plan-driven development lifecycle.

agile — Iterative and adaptive development principles.

iteration — Repeated short cycle yielding feedback/increment.

requirements — Functions/constraints a system must satisfy.

NEW LEARNING

Read, annotate and explain

Waterfall lifecycle

Sequential analysis, design, implementation, testing and maintenance.

Agile methods

Iterative delivery, feedback and reprioritisation in short cycles.

Choose on evidence

Stability, regulation, stakeholder access and change risk affect suitability.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

An air-traffic subsystem has carefully fixed specifications and requires formal approvals. Select a development approach.

Model answer

A plan-driven/waterfall approach can support documented phase gates and traceability; however, testing and validation must still occur throughout as appropriate. Safety assurance matters more than simply labelling it waterfall.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe the waterfall stages and one drawback of late requirement changes. [6]

Question 2

Explain two features of agile development in a public-services app. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate waterfall versus agile for replacing an exam-management system subject to fixed deadlines and changing rules. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Agile means a project needs no testing or documentation.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.